INDUSTRIAL INTERNSHIP REPORT

on

"Food Delivery Application"

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time. My project was a Food Delivery Application — a web-based platform that allows users to browse restaurants, view menus, add items to a cart, and place food orders online. The application was built using HTML, CSS, JavaScript, PHP, and MySQL. This internship gave me a very good opportunity to get exposure to industrial problems and design/implement solutions for that. It was an overall great experience.

In collaboration with

upskill Campus | UniConverge Technologies Pvt Ltd (UCT) | The IoT Academy

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1. Preface

This report documents my six-week industrial internship undertaken as part of the upskill Campus program in collaboration with UniConverge Technologies Pvt Ltd (UCT). During this period, I worked on developing a Food Delivery Application — a complete web-based platform built using HTML, CSS, JavaScript, PHP, and MySQL.

In today's digital era, online food ordering has become an essential service. The ability to design and develop such a platform from scratch provided me with hands-on experience in full-stack web development, database management, and software engineering practices.

My project focused on building a functional food delivery system with core features including user authentication, restaurant listing with cuisine filters, dynamic menu display, a shopping cart, and an order placement system — all connected to a live MySQL database.

The opportunity provided by upskill Campus and UCT gave me real industry exposure that goes far beyond classroom learning. The structured six-week program helped me understand how real-world software projects are planned, built, and tested.

I am deeply grateful to the entire upskill Campus and UCT team for this opportunity. I would also like to thank my college, St. Xavier's College, Ranchi, for encouraging participation in such programs. To my juniors and peers: I strongly recommend pursuing such internships — the practical learning you gain is invaluable for your career.

2. Introduction

2.1 About UniConverge Technologies Pvt Ltd

UniConverge Technologies Pvt Ltd (UCT) is a company established in 2013, working in the Digital Transformation domain and providing industrial solutions with prime focus on sustainability and Return on Investment (RoI).

For developing its products and solutions, UCT leverages various cutting-edge technologies including Internet of Things (IoT), Cyber Security, Cloud Computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, and Front-End development.

UCT's key products and platforms include:

- UCT Insight — An IoT platform for quick deployment of IoT applications, built with Java backend and ReactJS frontend, supporting MySQL and various NoSQL databases.
- Smart Factory Platform (Factory Watch) — A platform for smart factory needs providing scalable production and asset monitoring, OEE and predictive maintenance solutions.
- LoRaWAN-based Solutions — UCT is an early adopter of LoRaWAN technology, providing solutions in Agritech, Smart Cities, Industrial Monitoring, and Smart Metering.
- Predictive Maintenance — Industrial machine health monitoring leveraging Embedded Systems, Industrial IoT, and Machine Learning technologies.

2.2 About upskill Campus (USC)

upskill Campus, along with The IoT Academy and in association with UniConverge Technologies, has facilitated the smooth execution of the complete internship process. USC is a career development platform that delivers personalized executive coaching in a more affordable, scalable, and measurable way.

upskill Campus aims to upskill 1 million learners in the next 5 years by providing industry-relevant programs, internships, projects, interaction with industry experts, and career growth services.

Website: https://www.upskillcampus.com

2.3 Objective

The objectives of this internship program were:

- To gain practical experience of working in the industry.
- To solve real-world problems through technology.
- To improve job prospects through hands-on project experience.
- To develop a deeper understanding of the field and its applications.

- To foster personal growth including better communication and problem-solving skills.

2.4 Reference

[1] upskill Campus — https://www.upskillcampus.com

[2] UniConverge Technologies — https://www.uniconvergetech.in

[3] PHP Manual — https://www.php.net/manual/en/

[4] MySQL Documentation — https://dev.mysql.com/doc/

2.5 Glossary

Terms	Acronym
Hypertext Markup Language	HTML
Cascading Style Sheets	CSS
PHP: Hypertext Preprocessor	PHP
MySQL Database	MySQL
UniConverge Technologies	UCT
upskill Campus	USC

3. Problem Statement

In the assigned problem statement, the challenge was to design and develop a fully functional Food Delivery Web Application that enables users to order food online from multiple restaurants.

The key problems identified were:

- Lack of a centralized online platform where customers can browse multiple restaurants and their menus at one place.
- No streamlined process for users to add food items to a cart and place orders efficiently.
- Absence of a secure user authentication system to manage individual customer accounts and order histories.
- No restaurant-level menu management system categorizing food items by type (starters, main course, drinks, etc.).

The solution needed to be built using standard web technologies (HTML, CSS, JavaScript, PHP, MySQL) to demonstrate a complete full-stack implementation suitable for real-world deployment.

4. Existing and Proposed Solution

Existing Solutions & Their Limitations

Several food delivery platforms exist today such as Swiggy, Zomato, and Uber Eats. While these are feature-rich platforms, they have the following limitations for a learning and internship context:

- They are highly complex, commercial-grade systems with hundreds of microservices — not suitable for academic demonstration.
- They require mobile apps, GPS integration, payment gateways, and real-time logistics — far beyond a basic project scope.
- No open-source, lightweight version exists that covers the fundamentals in a simple, understandable way.

Proposed Solution

The proposed solution is QuickBite — a lightweight, fully functional food delivery web application that covers all core functionalities:

- User Registration and Login with secure password hashing (PHP password_hash/password_verify).
- Restaurant Listing page with search functionality and cuisine-based filters.
- Dynamic Menu Display grouped by food category for each restaurant.
- Shopping Cart with add/remove items and real-time total calculation.
- Order Placement with delivery address and payment method selection.
- Order History page showing all past orders and their status.

Code Submission (GitHub Link)

Code Link:
https://github.com/YOUR_USERNAME/upskillcampus/blob/main/FoodDeliveryApp.html

Report Submission (GitHub Link)

Report Link:
https://github.com/YOUR_USERNAME/upskillcampus/blob/main/FoodDeliveryApp_RiyaSingh_USC_UCT.pdf

5. Proposed Design / Model

The application follows a classic 3-tier architecture: Presentation Layer (HTML/CSS/JS), Business Logic Layer (PHP), and Data Layer (MySQL).

5.1 High Level Diagram

The high-level flow of the QuickBite application is as follows:

- User opens the web browser and visits the application URL.
- The frontend (HTML/CSS/JS) sends requests to the PHP backend via Fetch API.
- PHP processes the request, queries MySQL, and returns JSON responses.
- The frontend dynamically renders restaurants, menus, and order status from the JSON data.

```
[ User Browser ] → [ index.html + JS ] → [ PHP API ] → [ MySQL Database ]
```

5.2 Low Level Diagram

At a low level, the application is organized as follows:

- index.html — Single page application with multiple views (Home, Restaurant, Cart, Orders).
- php/auth.php — Handles login, register, logout actions with session management.
- php/restaurants.php — Returns restaurant list and individual restaurant menus.
- php/orders.php — Handles order placement and retrieval of order history.
- php/config.php — Database connection configuration and helper functions.
- sql/schema.sql — Database schema with 5 tables: users, restaurants, menu_items, orders, order_items.

5.3 Interfaces

The application provides the following user interfaces:

- Home Page — Hero section, restaurant grid with emoji/CSS visuals, search bar, cuisine filter chips.
- Restaurant / Menu Page — Restaurant info header, menu items grouped by category, cart sidebar.
- Login / Register Modal — Form-based authentication with validation and error messages.
- Checkout Modal — Order summary, delivery address input, payment method selection.

- Order History Page — List of all past orders with status badges and item details.
- Order Success Page — Confirmation screen with order ID and estimated delivery time.

6. Performance Test

Performance testing was conducted to ensure the application meets basic reliability and usability constraints for a functional web application.

6.1 Test Plan / Test Cases

#	Test Case	Expected Result	Status
1	User Registration	Account created, session started	**PASS**
2	User Login	Valid credentials grant access	**PASS**
3	Invalid Login	Error message displayed	**PASS**
4	Restaurant Listing	All 5 restaurants displayed	**PASS**
5	Search Filter	Results filtered by query	**PASS**
6	Cuisine Filter	Results filtered by cuisine type	**PASS**
7	Menu Display	Items grouped by category	**PASS**
8	Add to Cart	Item added, count updated	**PASS**
9	Quantity Update	Cart total recalculated	**PASS**
10	Order Placement	Order saved in database	**PASS**
11	Order History	Past orders displayed	**PASS**
12	Logout	Session destroyed, UI reset	**PASS**

6.2 Test Procedure

Testing was performed manually on a local development environment using XAMPP (Apache + MySQL + PHP) on Windows. The browser used was Google Chrome. Each test case was executed step by step and results were recorded.

- Database was populated using the schema.sql file with 5 restaurants and 21 menu items.
- Each user flow was tested end-to-end from registration through order placement.
- PHP API endpoints were tested independently by accessing them directly in the browser.
- Form validations were tested with both valid and invalid inputs.

6.3 Performance Outcome

All 12 test cases passed successfully. The application demonstrated reliable performance under local testing conditions:

- Page load time: Under 1 second on localhost.
- API response time: Under 200ms for all PHP endpoints.
- Database queries executed without errors across all 5 tables.
- Cart calculations were accurate across all tested combinations of items and quantities.

7. My Learnings

This six-week internship was a transformative learning experience. Here is a summary of my key learnings:

- Full-Stack Web Development: I built a complete web application from scratch — frontend, backend, and database — gaining end-to-end development experience.
- PHP Backend Development: I learned to build RESTful APIs using PHP, handling user sessions, form data, and JSON responses.
- MySQL Database Design: I designed a normalized relational database schema with foreign keys, joins, and transactions.
- Frontend JavaScript: I implemented dynamic UI updates without any external framework, using Fetch API for AJAX calls and DOM manipulation.
- Security Best Practices: I learned about password hashing using bcrypt, SQL injection prevention using prepared queries, and session management.
- Problem Solving: Debugging PHP errors, fixing MySQL connection issues, and resolving UI layout problems strengthened my analytical thinking.
- Project Structure: I understood how to organize a web project with clear separation of concerns — frontend, backend, and database layers.

These learnings have significantly improved my readiness for industry roles in web development. I now have a working project to showcase in interviews and a deeper understanding of how real-world applications are built.

8. Future Work Scope

Due to the six-week time constraint, several enhancements were identified that can be implemented in future iterations of this project:

- Real Payment Gateway Integration: Integrate Razorpay or PayPal to handle actual online payments instead of simulated payment methods.
- Admin Dashboard: Build a dedicated admin panel for restaurant owners to manage menus, view orders, and update order status in real time.
- Real-Time Order Tracking: Implement live order status updates using WebSockets or polling, showing delivery progress on a map.
- Mobile Responsive App: Convert the web application into a Progressive Web App (PWA) or build a native mobile app using React Native.
- Rating and Reviews: Allow users to rate restaurants and leave reviews after order delivery.
- Recommendation System: Use basic Machine Learning to recommend restaurants or dishes based on a user's order history.
- Email/SMS Notifications: Send order confirmation and delivery notifications via email (PHPMailer) or SMS (Twilio API).
- Multi-language Support: Add internationalization (i18n) support for Hindi and other regional languages.

These improvements would transform QuickBite from a functional prototype into a production-ready commercial food delivery platform.